INCH-POUND

MIL-M-38510/374(USAF)
AMENDMENT 2
17 June 1992
SUPERSEDING
AMENDMENT 1
27 June 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW POWER SCHOTTKY TTL, AND GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/374(USAF), dated 18 March 1983, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title; delete and substitute as printed above.

1.2.3, delete in its entirety and substitute:

"1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package"

1.3, delete and substitute:

"Thermal resistance, junction-to-case (Θ_{JC}):

Cases A, B, C, D	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/"

1.3, Junction temperature add "3/".

1.4, Recommended operating conditions, add "(V_{CC})" after Supply voltage.

Add footnotes 2/ and 3/ to bottom of page as follows:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883".

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PAGE 2

3.2.4, delete and substitute:

"3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

3.6, Marking: Delete the second sentence.

PAGE 3

* TABLE I, Output current, I_O, Min limits column: Delete "-30" and substitute "-20".

TABLE I, Propagation delay time, high to low level, t_{PHL}, Max limits column: Delete "12" and substitute "14".

PAGE 4

TABLE II, Interim electrical parameters: Delete "(pre burn-in)".

4.2a(1), delete "D or E" and substitute "D, E, or F".

4.2c, delete and substitute: "The percent defective allowable (PDA) shall be as specified in MIL-M-38510".

PAGE 5

4.4.3c(1), delete "D or E" and substitute "D, E, or F".

4.4.3c(3), delete "appendix B of MIL-M-38510" and substitute "method 1005 of MIL-STD-883".

6.1, delete in its entirety.

PAGE 7

FIGURE 1, delete "Cases X, Y" and substitute "Case 2". Make this change wherever cases X, Y appear.

PAGE 10

FIGURE 4, Note 1: Delete "=" after PRR.

PAGE 11

TABLE III, test number 49, pins 1Y, 2Y, delete "4.5 V". Also test number 49, pins 1A, 2A, add "4.5 V".

TABLE III, test number 50, pins 1Y, 2Y, delete "4.5 V". Also test number 50, pins 1A, 2A, add "GND".

PAGE 12

TABLE III, Subgroup 10, test number column: Delete "67-84" and "85-92".

TABLE III, Subgroup 10, t_{PHL}, Max limit column: Delete "12" and substitute "14".

* TABLE III, footnote 4/, delete in its entirety and substitute "4/ I_O limits shall be -20/-112 mA for circuit A and -30/-110 mA for circuit C."

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PAGE 13

TABLE III, V_{OH}, test number 3, pins 3A, 3B, 3C, delete "0.8 V" and substitute "2.0 V".

PAGE 14

TABLE III, Subgroup 10, test number column: Delete "72-80" and "81-89".

TABLE III, Subgroup 10, t_{PHL}, Max limits column: Delete "12" and substitute "14".

* TABLE III, footnote 4/, delete in its entirety and substitute "4/ I_O limits shall be -20/-112 mA for circuit A and -30/-110 mA for circuit C."

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-F725)

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